

Question:

Bank Account Management System:

Problem Statement: Design a system to manage bank accounts.

Classes: Account, Savings-Account, Current-Account.

Attributes: Account number, balance, account type, interest rate (for savings account),

overdraft limit (for current account), etc.

Methods: Deposit, Withdraw, Calculate Interest, Transfer, etc.

Object Usage: Create objects for each customer's

Steps For Execution:

1.Open your IDE (Eclipse/VS Code) or Command Prompt.

2.Create a new Java file named StudentUtilityProgram.java.

3.Write the complete program code inside the file.

Save the file.

4.Compile the program using: `javac StudentUtilityProgram.java`

5.Run the program using: `java StudentUtilityProgram`

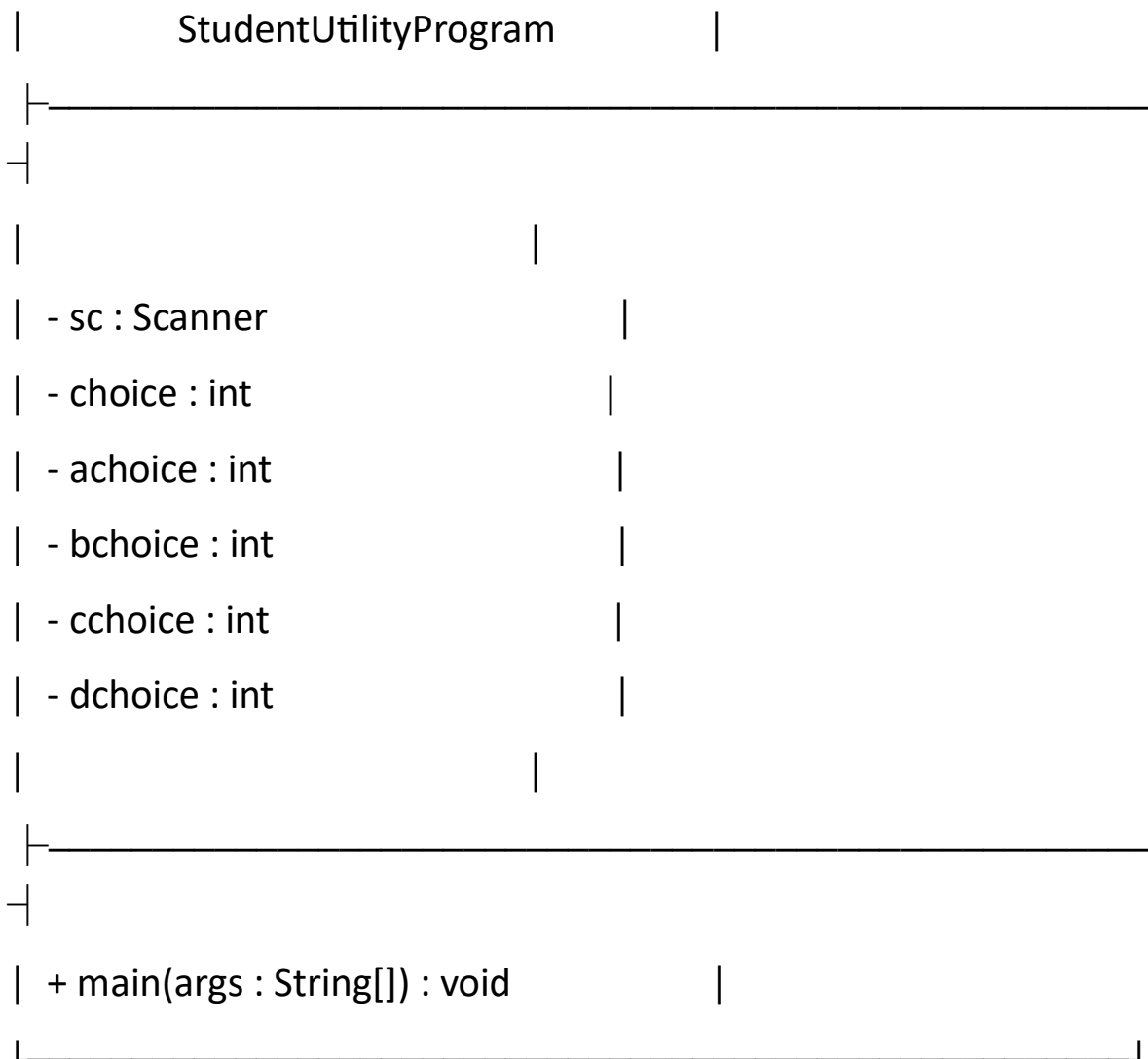
6.Enter the required choice and inputs as prompted.

7.Test all four parts and their respective menu options.

8.Observe the output for each selected option.

9.After completing the execution, close the program.

UML Diagram



uses



+ nextInt() : int	
+ nextDouble() : double	
+ nextLine() : String	
+ close() : void	

Code:

```
package javacore;
```

```
import java.util.Scanner;
```

```
public class StudentUtilityProgram {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("1. Part 1");
```

```
        System.out.println("2. Part 2");
```

```
        System.out.println("3. Part 3");
```

```
        System.out.println("4. Part 4");
```

```
        System.out.print("Enter your choice: ");
```

```
        int choice = sc.nextInt();
```

```
switch (choice) {
```

```
// ===== PART 1 =====
```

```
case 1:
```

```
    System.out.print("Enter case 1-4: ");
```

```
    int achoice = sc.nextInt();
```

```
    String name;
```

```
    int rollno;
```

```
    double marks1, marks2, marks3, total, percentage;
```

```
    switch (achoice) {
```

```
        // Case 1: Student Information
```

```
        case 1:
```

```
            System.out.println("Student Information");
```

```
            System.out.println("Name : Lalitha");
```

```
            System.out.println("Roll No : 101");
```

```
            break;
```

```
        // Case 2: Enter Student Details
```

```
        case 2:
```

```
            sc.nextLine(); // Clear newline
```

```
System.out.print("Enter Student Name: ");  
name = sc.nextLine();
```

```
System.out.print("Enter Roll Number: ");  
rollno = sc.nextInt();
```

```
System.out.print("Enter English Marks: ");  
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");  
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");  
marks3 = sc.nextDouble();
```

```
break;
```

```
// Case 3: Calculate Total and Percentage  
case 3:
```

```
System.out.print("Enter English Marks: ");  
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");
```

```
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");
```

```
marks3 = sc.nextDouble();
```

```
total = marks1 + marks2 + marks3;
```

```
percentage = (total / 300) * 100;
```

```
System.out.println("Total = " + total);
```

```
System.out.println("Percentage = " + percentage);
```

```
break;
```

```
// Case 4: Complete Student Result
```

```
case 4:
```

```
sc.nextLine(); // Clear newline
```

```
System.out.print("Enter Student Name: ");
```

```
name = sc.nextLine();
```

```
System.out.print("Enter Roll Number: ");
```

```
rollno = sc.nextInt();
```

```
System.out.print("Enter English Marks: ");
```

```
marks1 = sc.nextDouble();
```

```
System.out.print("Enter Maths Marks: ");  
marks2 = sc.nextDouble();
```

```
System.out.print("Enter Social Marks: ");  
marks3 = sc.nextDouble();
```

```
total = marks1 + marks2 + marks3;  
percentage = (total / 300) * 100;
```

```
System.out.println("\n----- RESULT -----");  
System.out.println("Name : " + name);  
System.out.println("Roll No : " + rollno);  
System.out.println("Total : " + total);  
System.out.println("Percentage : " + percentage);  
break;
```

```
default:
```

```
    System.out.println("Invalid Choice");
```

```
}
```

```
break;
```

```
// ===== PART 2 =====
```

case 2:

```
System.out.print("Enter case 1-4: ");
```

```
int bchoice = sc.nextInt();
```

```
switch (bchoice) {
```

```
    // Case 1: Even or Odd
```

```
    case 1:
```

```
        System.out.print("Enter a number: ");
```

```
        int n = sc.nextInt();
```

```
        if (n % 2 == 0)
```

```
            System.out.println("Even Number");
```

```
        else
```

```
            System.out.println("Odd Number");
```

```
        break;
```

```
    // Case 2: Greatest of Three Numbers
```

```
    case 2:
```

```
        System.out.print("Enter first number: ");
```

```
        int a = sc.nextInt();
```

```
        System.out.print("Enter second number: ");
```



```
int b = sc.nextInt();
```

```
System.out.print("Enter third number: ");
```

```
int c = sc.nextInt();
```

```
if (a >= b && a >= c)
```

```
    System.out.println("Largest = " + a);
```

```
else if (b >= a && b >= c)
```

```
    System.out.println("Largest = " + b);
```

```
else
```

```
    System.out.println("Largest = " + c);
```

```
break;
```

```
// Case 3: Grade
```

```
case 3:
```

```
System.out.print("Enter percentage: ");
```

```
double per = sc.nextDouble();
```

```
if (per >= 90)
```

```
    System.out.println("Grade A");
```

```
else if (per >= 75)
```

```
    System.out.println("Grade B");
```

```
else if (per >= 60)
```

```
        System.out.println("Grade C");  
    else if (per >= 40)
```

```
        System.out.println("Grade D");  
    else  
        System.out.println("Fail");
```

```
    break;
```

```
// Case 4: Day of the Week
```

```
case 4:
```

```
    System.out.print("Enter day number (1-7): ");  
    int day = sc.nextInt();
```

```
switch (day) {
```

```
    case 1:
```

```
        System.out.println("Monday");  
        break;
```

```
    case 2:
```

```
        System.out.println("Tuesday");  
        break;
```

```
    case 3:
```

```
        System.out.println("Wednesday");  
        break;
```

```
    case 4:
```

```
        System.out.println("Thursday");
        break;
    case 5:
        System.out.println("Friday");
        break;
    case 6:
        System.out.println("Saturday");
        break;
    case 7:
        System.out.println("Sunday");
        break;
    default:
        System.out.println("Invalid Day");
}
```

```
break;
```

```
default:
    System.out.println("Invalid Choice");
}
break;
```

```
// ===== PART 3 =====
```

case 3:

```
System.out.print("Enter case 1-4: ");
```

```
int cchoice = sc.nextInt();
```

```
switch (cchoice) {
```

```
    // Case 1: Multiplication Table
```

```
    case 1:
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        System.out.println("Multiplication Table:");
```

```
        for (int i = 1; i <= 10; i++) {
```

```
            System.out.println(num + " x " + i + " = " + (num * i));
```

```
        }
```

```
        break;
```

```
    // Case 2: Numbers from 1 to N
```

```
    case 2:
```

```
        System.out.print("Enter N: ");
```

```
        int n = sc.nextInt();
```

```
System.out.println("Numbers from 1 to " + n + ":");
```

```
for (int i = 1; i <= n; i++) {  
    System.out.print(i + " ");  
}
```

```
System.out.println();
```

```
break;
```

```
// Case 3: Sum of N Numbers
```

```
case 3:
```

```
System.out.print("Enter N: ");
```

```
int m = sc.nextInt();
```

```
int sum = 0;
```

```
for (int i = 1; i <= m; i++) {  
    sum = sum + i;  
}
```

```
System.out.println("Sum = " + sum);
```

```
break;
```

```
// Case 4: Fibonacci Series
```

case 4:

```
System.out.print("Enter number of terms: ");
```

```
int terms = sc.nextInt();
```

```
int a = 0;
```

```
int b = 1;
```

```
int c;
```

```
System.out.println("Fibonacci Series:");
```

```
for (int i = 1; i <= terms; i++) {
```

```
    System.out.print(a + " ");
```

```
    c = a + b;
```

```
    a = b;
```

```
    b = c;
```

```
}
```

```
System.out.println();
```

```
break;
```

default:

```
System.out.println("Invalid Choice");
```

```
}
```

```
break;
```

```
// ===== PART 4 =====
```

```
case 4:
```

```
    System.out.print("Enter case 1-4: ");
```

```
    int dchoice = sc.nextInt();
```

```
    switch (dchoice) {
```

```
        // Case 1: Factorial
```

```
        case 1:
```

```
            System.out.print("Enter a number: ");
```

```
            int n = sc.nextInt();
```

```
            int fact = 1;
```

```
            for (int i = 1; i <= n; i++) {
```

```
                fact = fact * i;
```

```
            }
```

```
            System.out.println("Factorial = " + fact);
```

```
            break;
```

```
// Case 2: Prime Number
```

```
case 2:
```

```
    System.out.print("Enter a number: ");
```

```
    int num = sc.nextInt();
```

```
    int count = 0;
```

```
    for (int i = 1; i <= num; i++) {
```

```
        if (num % i == 0) {
```

```
            count++;
```

```
        }
```

```
    }
```

```
    if (count == 2)
```

```
        System.out.println("Prime Number");
```

```
    else
```

```
        System.out.println("Not a Prime Number");
```

```
    break;
```

```
// Case 3: Maximum of Two Numbers
```

```
case 3:
```

```
    System.out.print("Enter first number: ");
```

```
    int a = sc.nextInt();
```



```
        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        if (a > b)
            System.out.println("Maximum = " + a);
        else
            System.out.println("Maximum = " + b);

        break;

// Case 4: Area of Circle
case 4:
    System.out.print("Enter radius: ");
    double r = sc.nextDouble();

    double area = 3.14 * r * r;

    System.out.println("Area of Circle = " + area);
    break;

default:
    System.out.println("Invalid Choice");
}
```

```
break;
```

```
default:
```

```
    System.out.println("Invalid Choice");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

1. Part 1

2. Part 2

3. Part 3

4. Part 4

Enter your choice: 1

Enter case 1-4: 1

Student Information

Name : Lalitha

Roll No : 101

Enter your choice: 1

Enter case 1-4: 2

Enter Student Name: Ramesh

Enter Roll Number: 101

Enter English Marks: 80

Enter Maths Marks: 90

Enter Social Marks: 85

Enter your choice: 1

Enter case 1-4: 3

Enter English Marks: 80

Enter Maths Marks: 90

Enter Social Marks: 85

Total = 255.0

Percentage = 85.0

Enter your choice: 1

Enter case 1-4: 4

Enter Student Name: Reshma

Enter Roll Number: 101

Enter English Marks: 80

Enter Maths Marks: 90

Enter Social Marks: 85

----- RESULT -----

Name : Reshma

Roll No : 101

Total : 255.0

Percentage : 85.0

Enter your choice: 2

Enter case 1-4: 1

Enter a number: 10

Even Number

Enter your choice: 2

Enter case 1-4: 2

Enter first number: 10

Enter second number: 25

Enter third number: 15

Largest = 25

Enter your choice: 2

Enter case 1-4: 3

Enter percentage: 85

Grade B

Enter your choice: 2

Enter case 1-4: 4

Enter day number (1-7): 3

Wednesday

Enter your choice: 3

Enter case 1-4: 1

Enter a number: 5

Multiplication Table:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

Enter your choice: 3

Enter case 1-4: 2

Enter N: 5

Numbers from 1 to 5:

1 2 3 4 5 1.

Enter your choice: 3

Enter case 1-4: 3

Enter N: 5

Sum = 15

Enter your choice: 3

Enter case 1-4: 4

Enter number of terms: 7

Fibonacci Series:

0 1 1 2 3 5 8

Enter your choice: 4

Enter case 1-4: 1

Enter a number: 5

Factorial = 120

Enter your choice: 4

Enter case 1-4: 2

Enter a number: 7

Prime Number

Enter your choice: 4

Enter case 1-4: 3

Enter first number: 25

Enter second number: 40

Maximum = 40

Enter your choice: 4

Enter case 1-4: 4

Enter radius: 5

Area of Circle = 78.5